Proof Observe that Nosé-Hoover-Langevin dynamics is a special case of the system (8.25)–(8.26) Applying the proposition in our case, the problem is to show that the Lie algebra generated by

$$\boldsymbol{F} = \begin{bmatrix} \boldsymbol{p} \\ -\boldsymbol{A}\boldsymbol{q} \end{bmatrix}, \qquad \boldsymbol{G} = \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{p} \end{bmatrix},$$

spans $\mathbb{R}^{2\mathrm{N_c}}$. We shall write $S\{\boldsymbol{F}, \boldsymbol{G}\}$ for the Lie algebra generated by $\boldsymbol{F}$ and $\boldsymbol{G}$.

Next we note:

Lemma 8.1 *For* $k = 1, 2, \ldots,$

$$\boldsymbol{C}_k = \begin{bmatrix} \boldsymbol{A}^{k-1}\boldsymbol{p} \\ \boldsymbol{A}^k\boldsymbol{q} \end{bmatrix} \in S\{\boldsymbol{F}, \boldsymbol{G}\},$$

and

$$\boldsymbol{D}_k = \begin{bmatrix} \boldsymbol{A}^k\boldsymbol{q} \\ -\boldsymbol{A}^k\boldsymbol{p} \end{bmatrix} \in S\{\boldsymbol{F}, \boldsymbol{G}\}.$$

The proof of this is straightforward by induction on k. (See exercises.)
The conclusion of the proof of the theorem is based on the following proposition:

Proposition 8.3 *Assume* $\boldsymbol{A}$ *to be a symmetric positive definite matrix with distinct eigenvalues, and let* $\boldsymbol{C}_k$, $\boldsymbol{D}_k$ *be defined as in Lemma 8.1, then the* $2\mathrm{N_c}$ *vectors* $\boldsymbol{C}_1, \boldsymbol{D}_1, \boldsymbol{C}_2, \boldsymbol{D}_2, \ldots, \boldsymbol{C}_{\mathrm{N_c}}, \boldsymbol{D}_{\mathrm{N_c}}$ *form a linearly independent set on* $\mathcal{D}$.

Proof $\boldsymbol{A}$, being symmetric positive definite is unitarily similar to a diagonal matrix $\boldsymbol{\Lambda}$, so $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^T$. Define $\hat{\boldsymbol{p}} = \boldsymbol{U}^T\boldsymbol{p}$, $\hat{\boldsymbol{q}} = \boldsymbol{U}^T\boldsymbol{q}$. The domain $\mathcal{D}$ is defined by the condition that $\hat{p}_i$ and $\hat{q}_i$ are not simultaneously zero.

With the change of variables, the problem is to show that $\hat{\boldsymbol{C}}_1, \hat{\boldsymbol{D}}_1, \ldots, \hat{\boldsymbol{C}}_{\mathrm{N_c}}, \hat{\boldsymbol{D}}_{\mathrm{N_c}}$ are linearly independent, where

$$\hat{\boldsymbol{C}}_k = \begin{bmatrix} \boldsymbol{\Lambda}^{k-1}\hat{\boldsymbol{p}} \\ \boldsymbol{\Lambda}^k\hat{\boldsymbol{q}} \end{bmatrix}, \qquad \hat{\boldsymbol{D}}_k = \begin{bmatrix} \boldsymbol{\Lambda}^k\hat{\boldsymbol{q}} \\ -\boldsymbol{\Lambda}^k\hat{\boldsymbol{p}} \end{bmatrix}.$$

Suppose the vectors are linearly dependent, then there exist coefficients $\alpha_0, \alpha_1, \ldots, \alpha_{\mathrm{N_c}-1}$ and $\beta_0, \beta_1, \ldots, \beta_{\mathrm{N_c}-1}$ not all zero such that:

$$\sum_{j=1}^{\mathrm{N_c}} \alpha_j \hat{\boldsymbol{C}}_j + \sum_{j=1}^{\mathrm{N_c}} \beta_j \hat{\boldsymbol{D}}_j = 0.$$